

Product Laws of Exponents*Question Bank - Additional 3.01*

Show your work. Circle your final answer.

For questions 1 and 2, write the product as a repeated multiplication expression, then as a single power.

1. $4^2 \cdot 4^5$

2. $7^2 \cdot 7^1 \cdot 7^3$

3. Evaluate each power, then multiply. $5^3 \cdot 5^2$

4. Apply the product of powers law, then evaluate the power. $(-6)^2 \cdot (-6)^7$

5. Leeland says he can evaluate $7^3 \cdot 3^7$ using the product of powers law. Is he correct? Explain.

For questions 6 through 8, use the power of a product law.

6. Write $(7 \cdot 8)^2$ as a repeated multiplication expression, group the repeated factors, then write it in the form $a^m \cdot b^n$.

7. Evaluate inside the parentheses first, then apply the exponent. $(-3 \cdot 6)^4$

8. Apply the power of a product law, then evaluate. $(2 \cdot (-9))^3$

For questions 9 and 10, use the power of a power law or the identity law.

9. Write $[(5)^4]^4$ as a single power, then give its value.

10. Rewrite $[(2)^5]^2$ using a law of exponents, name the law, then evaluate.

Working backwards.

11. Find the value of n that makes each statement true.

a. $3^n \cdot 3^4 = 3^9$

b. $(2^n)^3 = 2^{12}$

c. $\left(\frac{1}{2}\right)^n = \frac{1}{32}$

12. Which expression is equivalent to $(3^4)^2$?

- ☐ A. 3^8
- ☐ B. 3^6
- ☐ C. $2 \cdot 3^4$
- ☐ D. 3^{16}

13. Select ALL of the expressions that are equivalent to 10^6 .

☐ A. $10^3 \cdot 10^3$

☐ B. $(10^2)^3$

☐ C. $10^2 \cdot 10^4$

☐ D. 100^3

☐ E. $10^2 \cdot 10^3$

☐ F. $(10^3)^3$